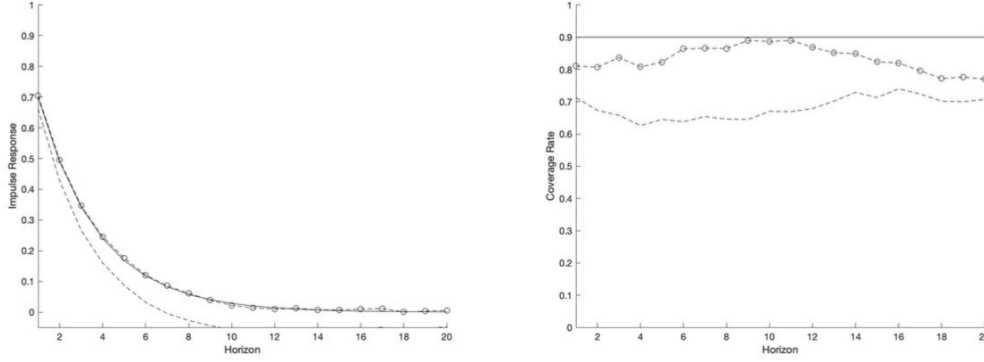
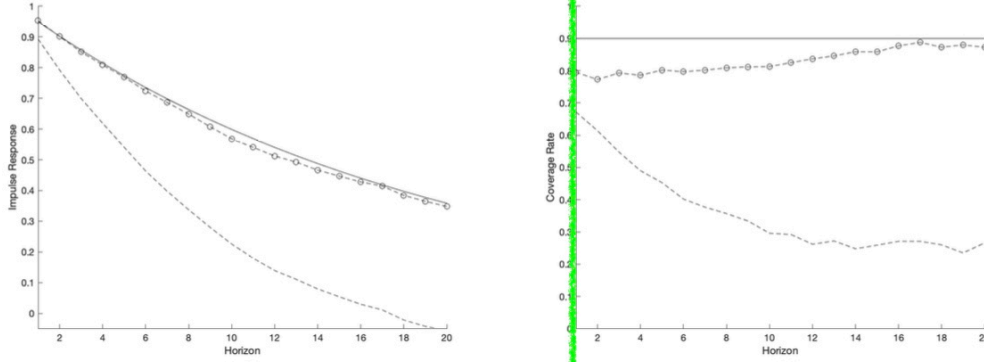


Figure 3
AR Model ($T = 100$)

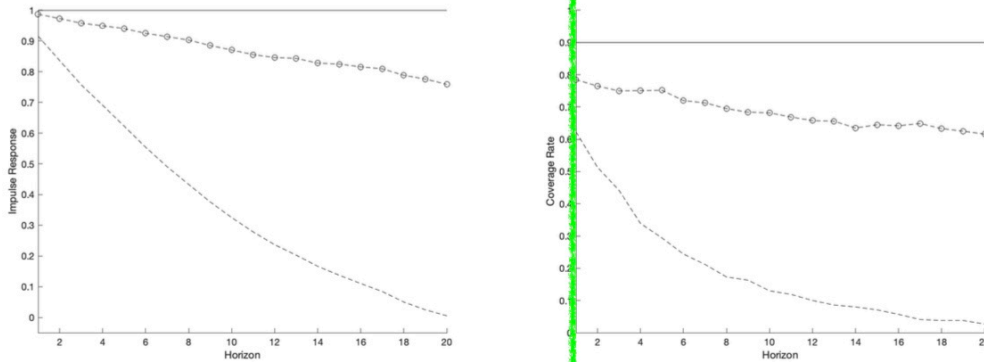
$$\phi = 0.7$$



$$\phi = 0.95$$



$$\phi = 1.0$$



Notes: This figure displays simulation results from estimation of the levels and differences specification of the LP when the true DGP is an AR(1) model and $T = 100$. Results for three alternative values of the autoregressive parameter ($\phi = \{0.7, 0.95, 1.0\}$) are displayed. The left column shows the average impulse response function estimate for the levels specification (dashed line) and differences specification (dash-circle line), as well as the true impulse response function (solid line). The right column shows the 90% confidence interval coverage of the true impulse response function for the levels specification (solid line) and differences specification (dash-circle line).

Figure 1: LP bias in SMALL samples, $T = 100, \rho = 0.95$ or $1, Nreps = 10,000$

True DGP: $y_t = \rho y_{t-1} + u_t, \quad u \sim N(0, 1), \quad \rho = 0.95 \text{ or } 1, \quad T = 100$

Estimated OLS models:

Intercept: no $y_t = \beta_1 y_{t-1} + u_t$

Intercept: yes $y_t = \beta_1 y_{t-1} + \beta_0 + u_t$

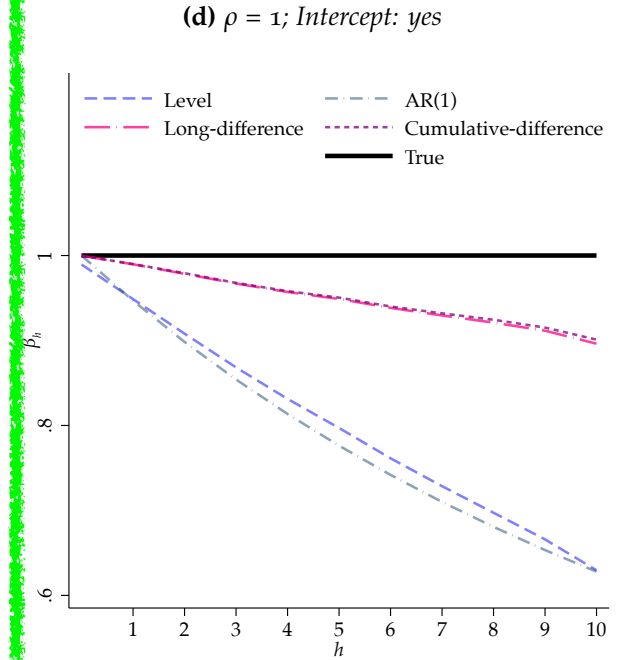
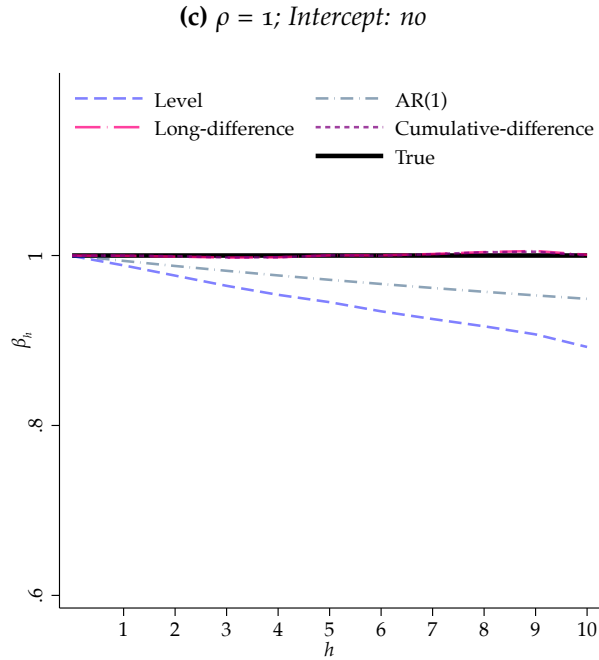
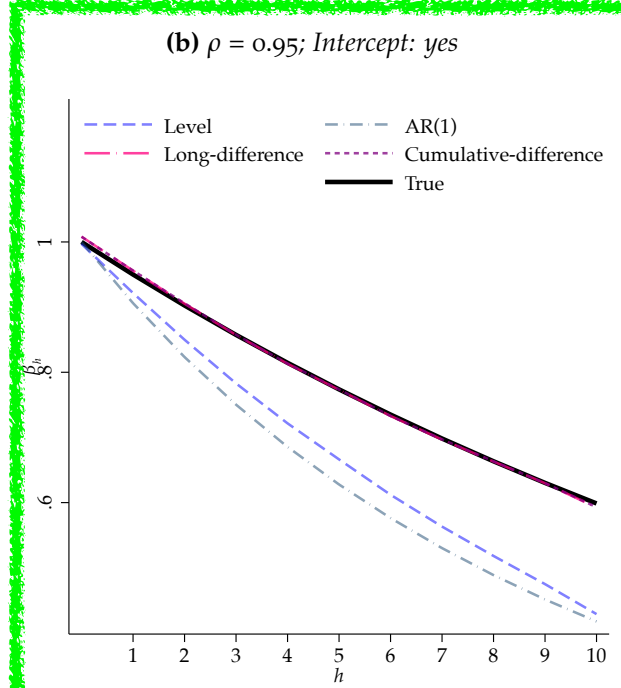
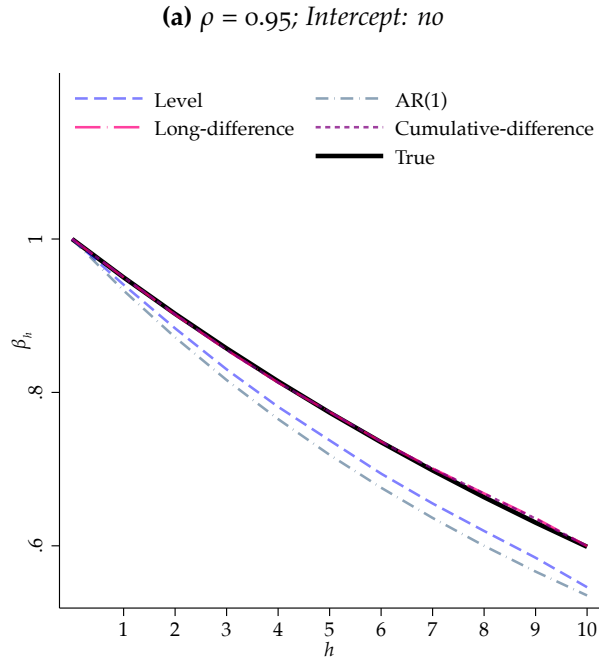


Figure 2: LP bias in LARGE samples, $T = 100,000$, $\rho = 0.95$ or 1 , $Nreps = 100$

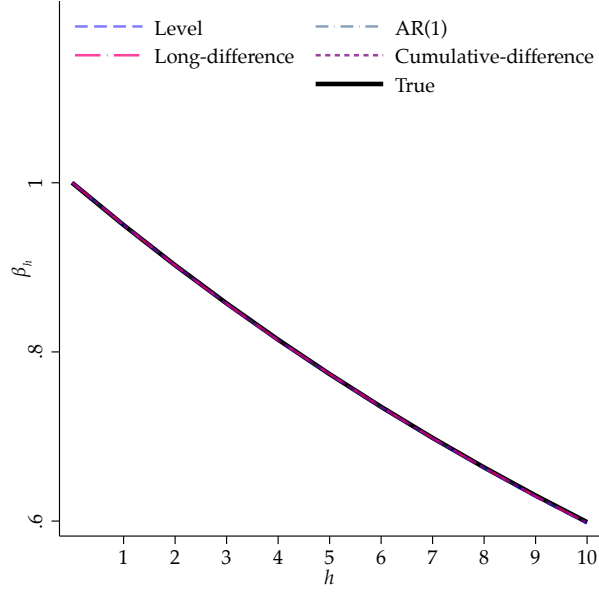
True DGP: $y_t = \rho y_{t-1} + u_t$, $u \sim N(0, 1)$, $\rho = 0.95$ or 1 , $T = 100,000$

Estimated OLS models:

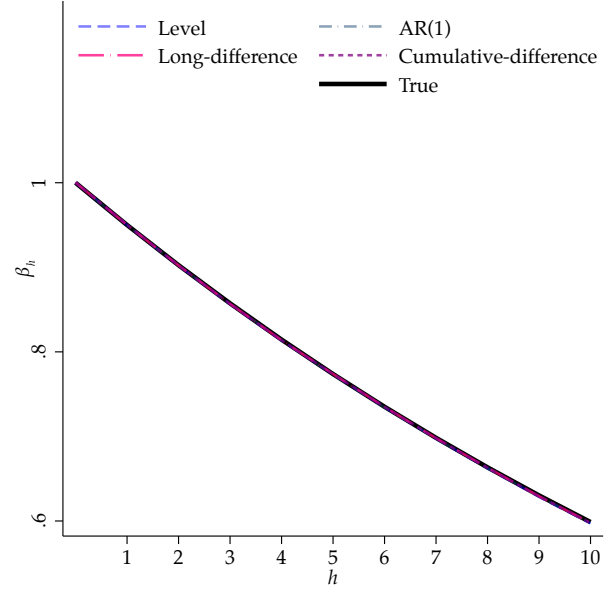
Intercept: no $y_t = \beta_1 y_{t-1} + u_t$

Intercept: yes $y_t = \beta_1 y_{t-1} + \beta_0 + u_t$

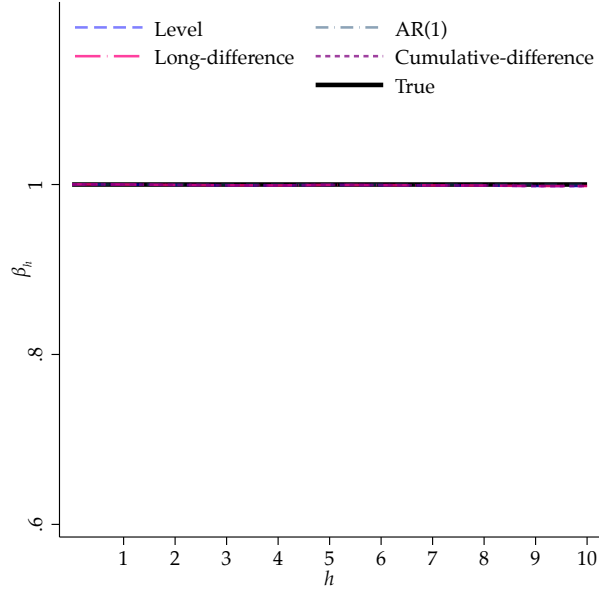
(a) $\rho = 0.95$; Intercept: no



(b) $\rho = 0.95$; Intercept: yes



(c) $\rho = 1$; Intercept: no



(d) $\rho = 1$; Intercept: yes

